

Development of CaSO₄: Li Composites by Different Synthesis Routes for Applications in TL and OSL Dosimetry

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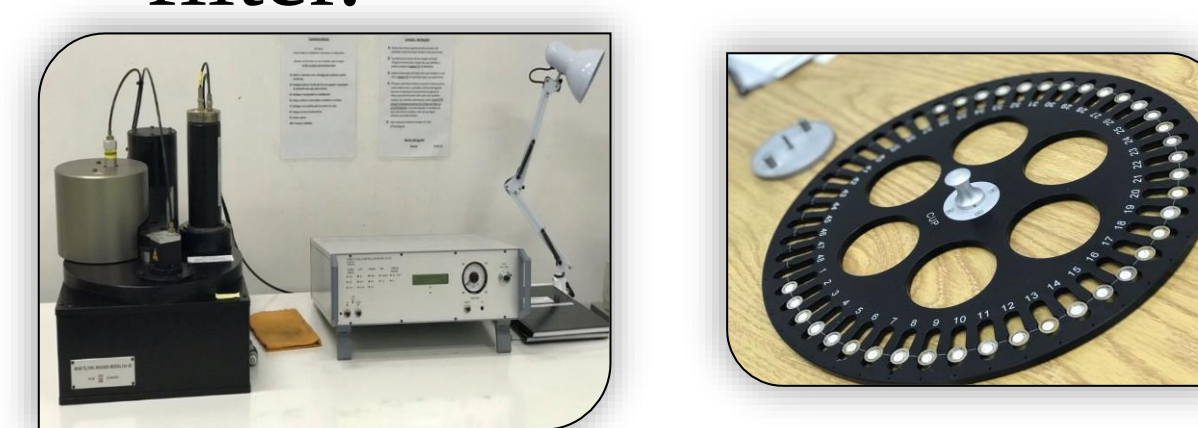
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INTRODUCTION

Lithium (Li) has gained prominence as an alternative dopant in the development of luminescent materials due to its promising optical properties[1–2]. It presents a viable option compared to rare earth elements, which, although efficient, are expensive, less abundant, and exhibit limitations related to energy levels. Furthermore, Li is abundant, affordable, and chemically versatile, capable of acting both as a luminescence activator and as a charge compensator, reinforcing its potential as a dopant[1]. However, the effects of Li monodoping in calcium sulfate (CaSO₄) matrices remain scarcely explored in the literature. In this context, this work aims to investigate the effect of Li as a sole dopant in the CaSO₄ matrix, evaluating its luminescent properties through thermoluminescence (TL) and optically stimulated luminescence (OSL) techniques.

MATERIALS AND METHODS

Solid-state diffusion (SSD) [3]	Chemical precipitation (PRE) [4]	Slow evaporation (SER) [5]	Structural characterization	Luminescent characterization
CaSO ₄ ·2H ₂ O+ Li ₂ CO ₃ + deionized water	Ca(C ₂ H ₃ O ₂) ₂ + Li ₂ CO ₃ + H ₂ O, mixed (NH ₄) ₂ SO ₄ in ethanol	CaCO ₃ +H ₂ SO ₄ + Li ₂ CO ₃	 DRX MEV	• ⁹⁰Sr/⁹⁰Y Beta Source; • Absorbed dose rate of 4.33Gy/min; • TL analyses: heating rate of 5°C/s was used; • OSL measurements: 470 nm blue LEDs; • TL and OSL using Hoya U-340 (340 ± 40) nm filter. 
Dried in an oven at 100°C	Filtration of CaSO ₄ , washing, and drying in an oven at 100°C	Heat to 375°C until all the acid has evaporated Washing Hot (100°C) Cold (25°C)		
Calcination 600°C for 1h	Calcination 600°C for 1h	Calcination 600°C for 1h		
Powder + Teflon 1:1 >>> Uniaxial pressure 200 kgf/1min >>> Sinterization 400 °C/1h				

RESULTS

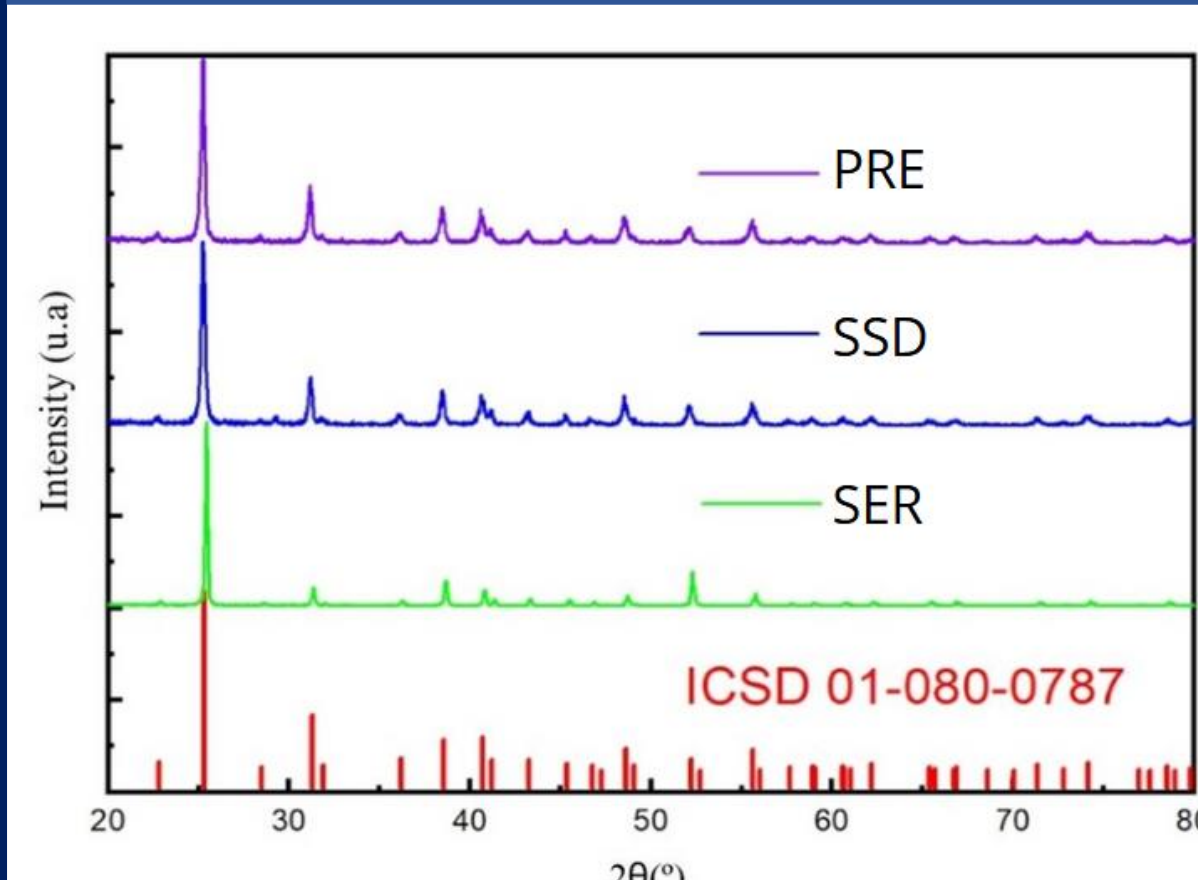


Fig. 1: X-ray diffraction (XRD) patterns of CaSO₄:Li powders synthesized via PRE, SSD and SER routes.

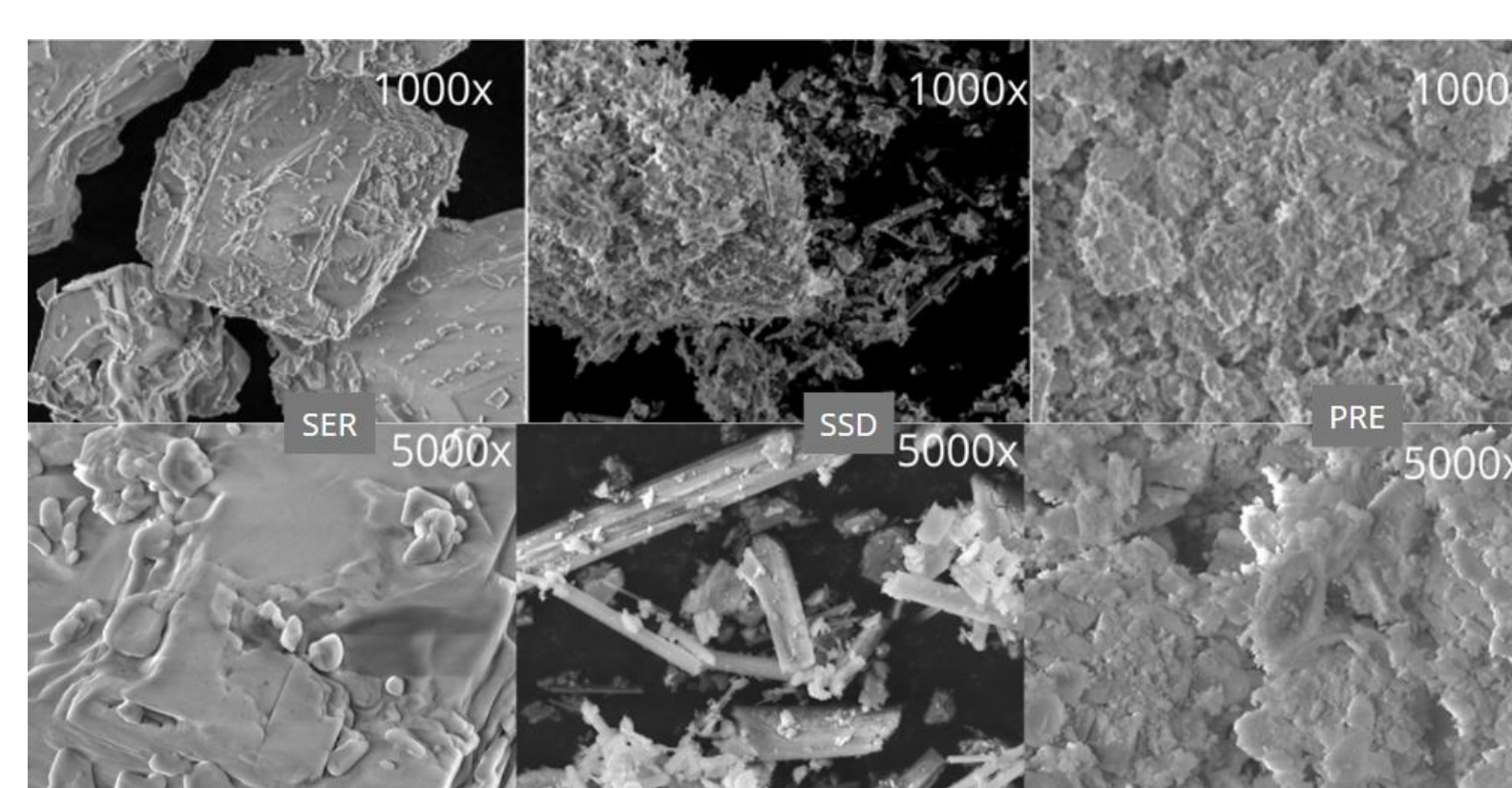


Fig. 2: SEM micrographs of CaSO₄:Li polycrystals synthesized via PRE, SSD and SER routes.

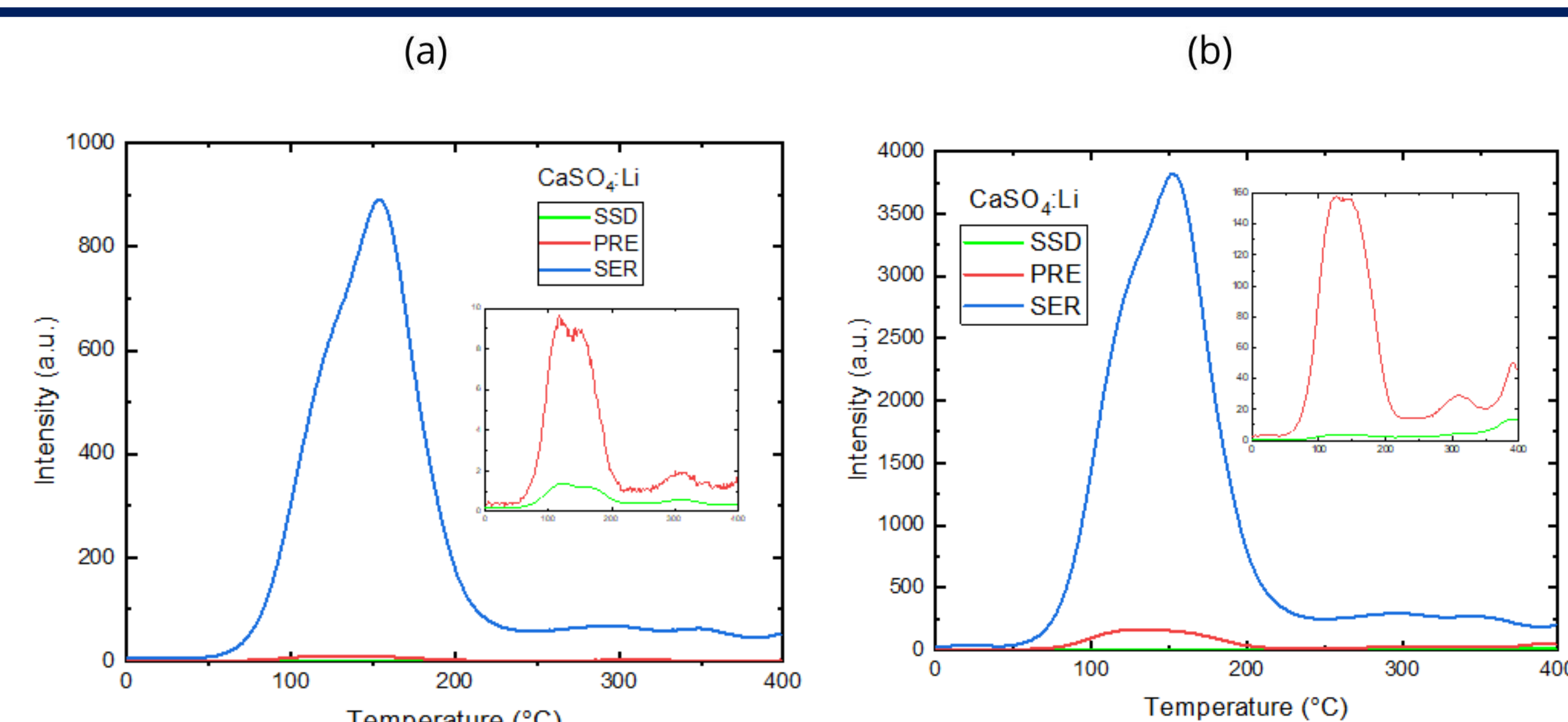


Fig. 3: TL glow curves of CaSO₄:Li samples measured using (a) Hoya U-340 and (b) Schott BG-39 filters for the PRE, SSD and SER routes.

CONCLUSION

- CaSO₄:Li phosphors synthesized via slow evaporation exhibited a single orthorhombic phase with large, irregular, well-defined crystallites.
- A main TL peak around ~150 °C was observed for all synthesis routes, with slow evaporation showing the highest luminescence efficiency.
- OSL decay curves displayed fast exponential decay, indicating trapping centers with high photoionization cross-section.
- Slow evaporation samples showed the highest OSL sensitivity, followed by solid-state reaction and co-precipitation.
- Results highlight slow evaporation as the most effective route for producing high-performance TL and OSL phosphors.

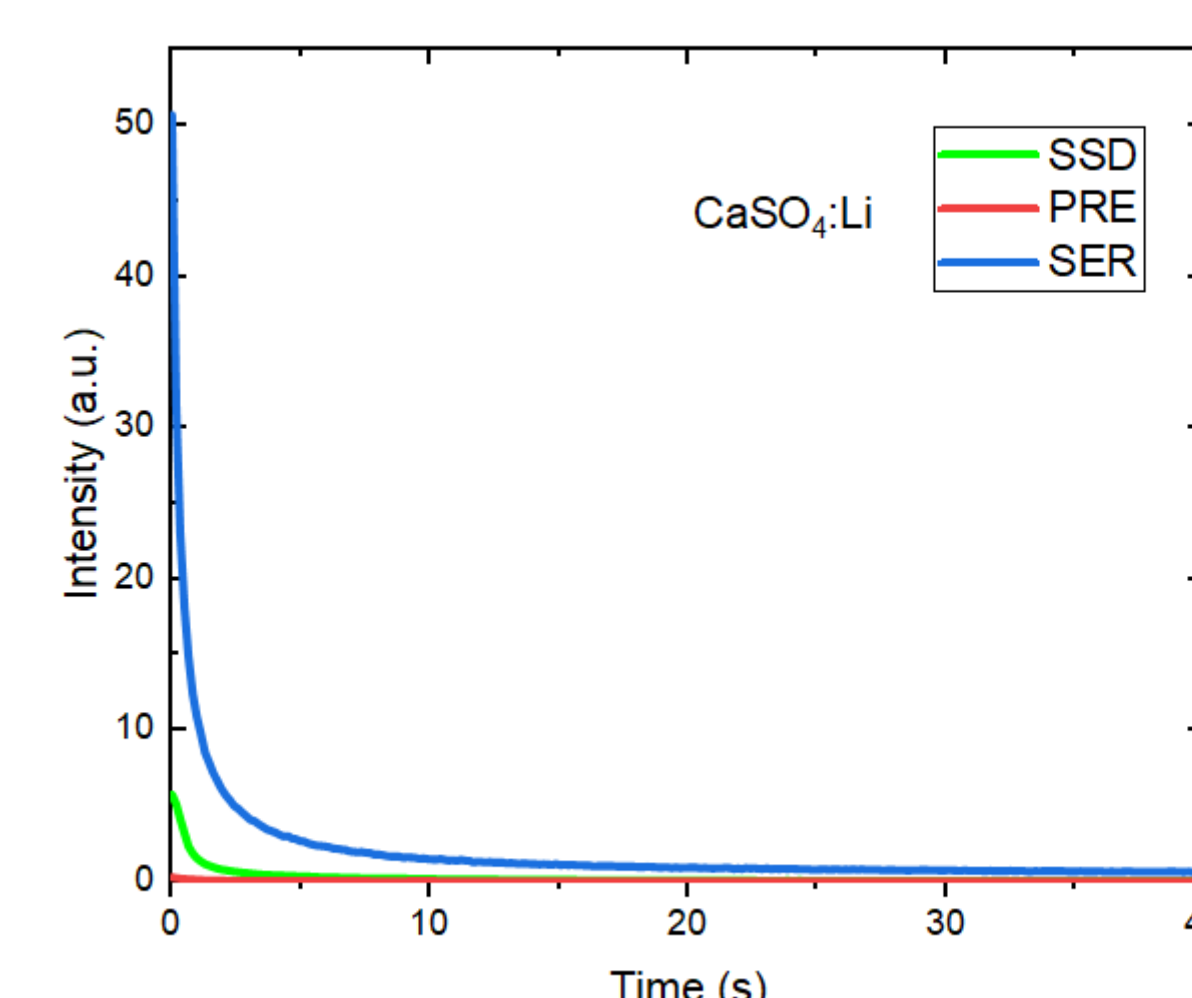


Fig. 4: OSL decay curves of CaSO₄:Li samples synthesized via PRE, SSD and SER routes.

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References

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